

Definition 3.12 (Group isomorphism). A group isomorphism f is said to be an isomorphism when it is bijective.

If there exist a group isomorphism between two groups, this establishes an equivalence relation (because the identity map is an isomorphism, isomorphisms can be inverted, and isomorphisms can be composed): we say that the two groups are isomorphic to each other, and we denote it with $\cong$.

Check precisely why the inverse is guaranteed to be an isomorphism.

3.5 Smallest finite groups

The smallest finite group is the trivial group $\mathbb{Z}_1 = \{e\}$, because to satisfy the requirement about the unit element a group must have at least one element. We also studied in detail $\mathbb{Z}_2 = \{e, a\}$, and we saw that we are constrained in the definition of its group product.

Now, we can consider a group of order 3, which in principle is $\{e, a, b\}$. Of course, we are constrained in defining all products involving e , but we have three possible choices for $a \cdot b$. If we put

$$a \cdot b = b \implies a \cdot e = e$$

which is absurd, and also

$$a \cdot b = a \implies e \cdot b = e,$$

thus it must be $a \cdot b = e$. The same goes also for $b \cdot a = e$, and similar arguments lead us to $a \cdot a = b$ and $b \cdot b = a$. Also in this case, we conclude that there is only one group with three elements: $\mathbb{Z}_3$.

Definition 3.13 (Cyclic group). For every positive integer n , the cyclic group $\mathbb{Z}_n$ is defined as

$$\{e, a, a^2, \dots, a^{n-1}\}.$$

Remark 3.14 (Roots of unity). One realisation of the cyclic groups is given by the complex n th roots of 1

$$\mathbb{Z}_n = \{\exp(2\pi i k/n) \text{ for } 0 \leq k < n\}$$

where the group product is the multiplication between complex numbers.

4 Lesson 4

4.1 Again, smallest finite groups

We will continue talking about small groups.

Definition 4.1 (Order of a group). For a finite group G , the order of the group $|G|$ is defined as the number of elements inside of it.

Definition 4.2 (Order of an element). The order of $a \in G$ (and $a \neq e$) is the smallest n such that $a^n = e$.

At order four, there are two different groups:

1. the cyclic group $\mathbb{Z}_4$, where the group product is defined analogously with addition modulo 4,
2. the product $\mathbb{Z}_2 \times \mathbb{Z}_2$, which is called the viergruppe V_4 .

Then, at order five, there is only $\mathbb{Z}_5$. Instead, at order six, in addition to $\mathbb{Z}_6$, we also have the group of symmetries of the equilateral triangle D_3 . It turns out that its six elements can be defined, equivalently, as

$$\{e, a, b, aba, ab, ba\}$$

and the operation between them such that

$$a^2 = b^2 = e \quad (ab)^3 = (ba)^3,$$

which implies that a , b and aba are reflections, while e , ab and ba the rotations. Looking at its Cayley table it is clear that it is not abelian: we already stated this property when we defined it through rotations, and reflections.

4.2 Symmetric groups and permutations

D_3 can also be denoted by S_3 , when identified with the symmetric group of degree 3: the permutations of 3 elements/objects.¹

Definition 4.3 (Permutation). Given a set X , such that $X \neq \emptyset$, a permutation (of the elements of X) is a bijective function

$$f : X \rightarrow X,$$

or equivalently $f \in \text{Map}(X, X)$.

The set of permutations of X is denoted as $\text{Perm}(X)$, and it is a group if we define the group product as function composition.

Of course, if $|X| = N$

$$|\text{Perm}(X)| = N!,$$

and Perm is also denoted as S_N .

For example, $S_1 = \mathbb{Z}_1$, but also $S_2 = \mathbb{Z}_2$.

Permutations can be represented in two ways:

1. in terms of a matrix with two rows and N columns

$$P = \begin{pmatrix} 1 & 2 & \dots & N \\ x_1 & x_2 & \dots & x_N \end{pmatrix}$$

where $f(i) = x_i$,

2. in terms of cycles, for example

$$1 \rightarrow 3 \rightarrow 2 = (132) \quad \text{is equivalent to} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

Remark 4.4. It is not guaranteed that a cycle contains all of the elements of the permutations, for example

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \longrightarrow 1 \rightarrow 2 = (12).$$

¹The permutations have a one to one correspondence with the symmetries of the triangle: for other polygons this is not true.

Remark 4.5. Distinct cycles commute

$$(14)(23) = (23)(14),$$

and positive permutations of the cycles are equivalent

$$(132) = (321) = (213)$$

even if by convention we always start with the smallest element.

This notation is particularly useful to find the inverse of an element: it suffices to read the sequence backwards. For example, if

$$P = (132) \implies P^{-1} = (231) = (123).$$

It is also possible to represent the group using cycles of length two with repeated elements (if necessary). For example

$$P = (132) = (13)(32).$$

Definition 4.6 (Signature). The signature of a permutation is defined as

$$\text{sgn}(P) = \begin{cases} +1 & \text{if } P \text{ is written as a product of an even number of permutations of length 2} \\ -1 & \text{if } P \text{ is the product of an odd number of two-cycles.} \end{cases}$$

It turns out that the even permutations form a subgroup, and it is called the alternating group A_N , and

$$|A_N| = \frac{|S_N|}{2} = \frac{N!}{2}.$$

Theorem 4.7 (Cayley)

Every finite group of order N is isomorphic to some subgroup of S_N .

Proof. Let's assume that G has order N , and consider the set

$$K = \{\forall g \in G : f_g : x \mapsto g \cdot x\},$$

which contains N bijective² functions.

Then, K is a group if we take the composition of function as the group product.

The mapping $\mathcal{H} : G \rightarrow K$ which maps $g \mapsto f_g$ is a group isomorphism. Finally, f_g are permutations: it is a subgroup of the permutations which is isomorphic to G . We took G as an arbitrary, thus the theorem has been proven. $\square$

It is also possible to define permutations of a multiset: a set in which elements can be repeated. It suffices to use the multinomial coefficient, for example

$$Y = \{1, 2, 2, 3, 3\} \longrightarrow \frac{5!}{1!2!2!}$$

where we divide $N!$ by the factorials of the multiplicities of the elements in the multiset.

²Because $\forall g \in G$, we have that $f_g^{-1} : (g \cdot x) \mapsto g^{-1}(g \cdot x) = x$.

4.3 Partitions and Young diagrams

For every permutation we have that

$$\sum n_i = N$$

where n_i is the length of the i th cycle. This means that $n_1, n_2, \dots$ are a *partition* of N .

To compute the number of partitions $p(N)$, we can construct the generating function

$$\sum_{N=0}^{\infty} p(N)x^N = \prod_{k=1}^{\infty} \frac{1}{1-x^k}.$$

Each partition can be associated with a *Young diagram*: an array of N boxes, in which the rows have non-increasing length. Each row corresponds to a number in the partition: order does not matter, because it is fixed by the non-increasing condition.

These Young diagrams will be useful when talking about different representations of groups (relevant in physics).

4.4 Free groups

Definition 4.8 (Free group). Given a group G and a subset $X \subset G$, when $\forall g \in G$ (but $g \neq e$)

$$g = x_{j_1}^{i_1} x_{j_2}^{i_2} \cdots x_{j_m}^{i_m}$$

with $X = \{x_1, x_2, \dots\}$ (that does not contain e), $x_a \neq x_{a+1}$ and $i_1, i_2, \dots$ (non-zero) integer exponents, then the elements x_i are called free generators of G (or “letters”), and G is called a free group $F(x)$.

A generic product of letters is called a ‘*word*’, and a word with non-zero i_1, i_2 and $x_a \neq x_{a+1}$ is a *reduced* word.

Remark 4.9. There can be groups for which there exist some words such that

$$x_{j_1}^{i_1} x_{j_2}^{i_2} \cdots x_{j_m}^{i_m} = e,$$

and in this case this equation is known as *relation*.

For example, for $\mathbb{Z}_N$ we have that $X = \{a\}$, but

$$a^N = e.$$

Conventionally, a group can be denoted with its *presentation*

$$G = \langle x_1, x_2, \dots, x_N | r_1, r_2, \dots, r_N \rangle,$$

for example $\mathbb{Z}_N = \langle a | a^N \rangle$, or also $\mathbb{Z} \times \mathbb{Z} = \langle a, b | aba^{-1}b^{-1} \rangle$.³

D_4 is another relevant example, which can be generated with just two letters

$$D_4 = \langle r, f | r^4, f^2, rfrf \rangle$$

where r is a rotation by $\pi/2$ and f is the reflection along the vertical axis.

³In the second example the relation is equivalent to asserting that $ab = ba$.

The Pauli group G_1 is relevant in quantum information theory, where

$$G_1 = \{\pm \mathbb{I}, \pm i\mathbb{I}, \pm \sigma_a, \pm i\sigma_a\},$$

and its presentation is

$$G_1 = \langle \sigma_1, \sigma_2, \sigma_3 \mid \sigma_a^2 = \mathbb{I}, \sigma_a \sigma_b = i\epsilon_{abc} \sigma_c \rangle$$

where the second relation holds for $a \neq b$.

4.5 Braid group

We can operate moves between pairs of strands, the most simple one is the Reidemeister move of the second type: undoing a passover.

Interchanging the strand i with $i + 1$ (with i on the front) is denoted by σ_i . Its inverse is different (as we see with the Reidemeister move of the second type), and it is called σ_i^{-1} .